

strong association between training and the emergence of non-productive tasks helps contextualize the finding in Section D.2.3 that training—when implemented in isolation—is linked to consistently lower productivity gains from AI chatbots.

D.3 Total Time Savings

What is the economic significance of the reported benefits of using AI chatbots? In Figure D.6, we combine workers’ frequency of use (from Figure D.3) with their reported time savings per day of usage (from Table D.2) to estimate time savings as a percent of total work hours.

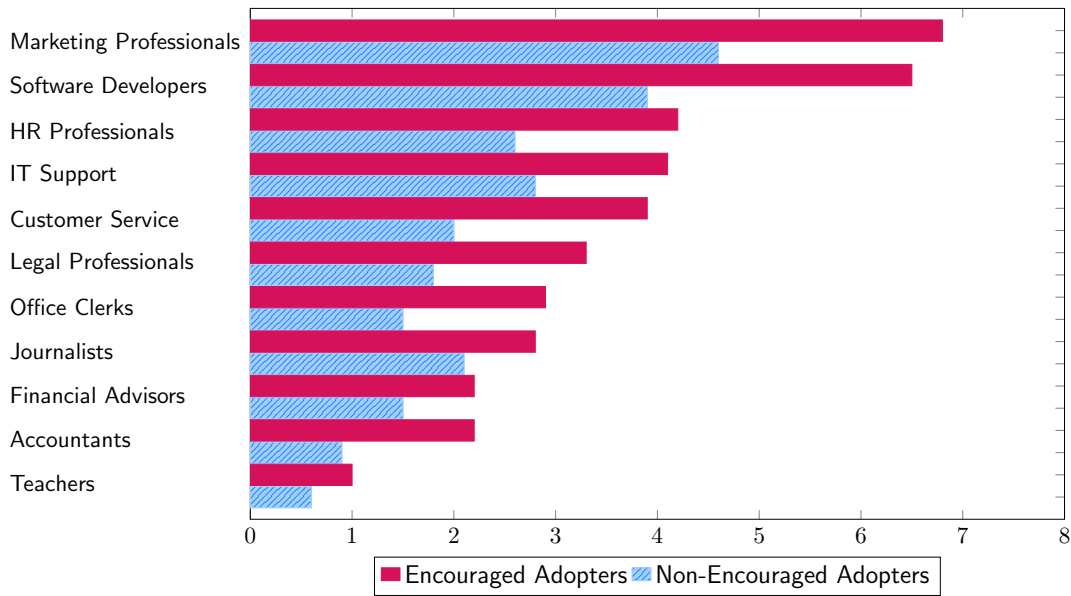
While time savings capture only one dimension of chatbot benefits—arguably the least transformative—their clear measurability in minutes per day makes them easy to quantify and compare. Moreover, because prior RCTs have also used time savings as a key metric for productivity gains, our estimates can be benchmarked against those studies. Finally, time savings remain the most commonly reported benefit among workers in our survey.³⁶

Figure D.6 shows estimated time savings by workers’ occupations and whether their employer encourages chatbot use. We focus on the encouraged-use policy, as Sections D.2.2 and D.2.3 identify it as the primary driver of increased adoption and reported benefits. The occupational breakdown facilitates comparison with prior studies focused on specific job roles.

Average time savings from AI chatbots vary from 6.8% among marketing professionals whose employers encourage their use to 0.6% among teachers in schools that do not encourage their use. Across our 11 occupations and employer policies, the average time savings amount to 2.8% of the total work hours of users.

³⁶Section 3.1 examines how various AI chatbot benefits—including time savings, enhanced creativity, improved quality, and other work-related dimensions—affect workers’ earnings and hours.

Figure D.6: Total Time Savings from AI Chatbots in % of Work Hours



Notes: This figure presents workers' time savings from AI chatbots, categorized by workers' occupations and employer policies. The figure focuses on workers who have ever used AI chatbots for work. Occupations are sorted by their average time savings among encouraged users. We estimate total time savings by combining workers' frequency of use (from Figure D.3) with their reported time savings per day of usage (from Table D.2). Table E.9 decomposes the time savings into each of its components: frequency of use, time savings per day of usage, and the covariance between the two. The table notes provide additional details on how time savings and usage frequencies are coded. *Sample:* The figure is based on all completed responses from the 2024 survey that can be linked to the registry data.

D.4 Comparison to the Literature

The widespread adoption of AI chatbots—and the substantial gender gaps in usage—are well documented in the literature.³⁷ We contribute to this evidence by documenting that employer initiatives for AI chatbot adoption are now common and play an important role in shaping overall adoption rates and demographic disparities in usage.³⁸

The importance of employer policies in driving adoption aligns with Dillon et al. (2025), who show that firm effects are a key determinant of workers' uptake of a randomized offer for Microsoft Copilot. The importance of training in closing gender gaps in adoption is

³⁷For example, our 2023 survey found that 39.8% of workers had used ChatGPT for work, with adoption rates 15.9 percentage points lower among women within occupations (Humlum and Vestergaard, 2025) (see Appendix E.1.4 for a comparison of the 2023 and 2024 adoption patterns). Higher adoption rates among younger, less experienced, and higher-earning workers also replicate the patterns observed in our earlier round. Bick, Blandin and Deming (2025) report similar adoption patterns among U.S. workers, and Otis et al. (2024) show that gender gaps in Generative AI adoption are pervasive across contexts.

³⁸By contrast, earlier surveys found relatively low take-up rates of AI at the firm level. For example, Bonney et al. (2024) document that as of February 2024, only 5.4% of U.S. firms reported using AI, with adoption in the leading sector (Information) still below 20%.

consistent with our 2023 survey round, in which women were significantly more likely to cite lack of training as a barrier to using ChatGPT (Humlum and Vestergaard, 2025).

More broadly, the role of employer initiatives in AI chatbot adoption echoes a large literature on the diffusion of general-purpose technologies, which highlights the need for complementary investments and coinvention (Bresnahan and Greenstein, 1996; David, 1990; Feigenbaum and Gross, 2024). The fact that initiatives can be counterproductive in isolation, yet complementary in combination, aligns with the concept of “Productivity J-curves,” where firm-based complementary investments are needed to unlock the potential of new technologies (Brynjolfsson, Rock and Syverson, 2021; David, 1990).

The reported time savings in Section D.3 may seem small compared to the substantial productivity gains—often exceeding 15%—documented by RCTs in our study occupations.^{39,40,41,42} How can we reconcile workers’ modest reported time savings with the substantial effects observed in RCTs?

First, as Figure D.6 shows, our reported time savings are highest precisely in the occupations covered by RCTs, such as customer and IT support (Brynjolfsson, Li and Raymond, 2025), marketing and human resources (Noy and Zhang, 2023), and software development (Peng et al., 2023). In contrast, reported savings are about half as large

³⁹Noy and Zhang (2023) report time savings of 37% from ChatGPT in text writing tasks representative of marketing and HR professionals. Brynjolfsson, Li and Raymond (2025) estimate productivity effects of 15% from a GPT-based chat assistant among customer and IT support agents, with more pronounced effects for less experienced workers. Cui et al. (2024) estimate a 26% productivity gain from GitHub Copilot among software developers, while Peng et al. (2023) report a 58% increase.

⁴⁰Our estimated productivity gains are closer to those of Acemoglu (2025), though for different reasons. While Acemoglu (2025) combines the experimental productivity estimates with a calibrated adoption curve, we measure both adoption and (self-reported) time savings directly. Bick, Blandin and Deming (2025) provide a thoughtful comparison between their survey-based estimates of productivity gains and the predictions of Acemoglu (2025).

⁴¹Our time savings estimates align more closely with those of Bick, Blandin and Deming (2025), who, in a nationally representative U.S. survey, find an average reported time savings of 5.4% among users. However, their study does not split the time savings by occupation or employer policies, limiting direct comparability to our estimates.

⁴²Edelman, Ngwe and Peng (2023) shows that self-reported time savings from AI chatbots—particularly Microsoft Copilot—may overstate actual productivity gains. This further reinforces the point that workers’ productivity improvements from AI chatbots are modest, as our self-reported time savings estimates would then represent an upper bound on realized benefits. That said, overstated gains may not be the full story: Section 4.1 shows that workers’ self-reported earnings effects closely align with our difference-in-differences estimates based on administrative data, reflecting that workers recognize that AI chatbots have not materially improved their earnings.

among teachers, accountants, and financial advisors, whose tasks may be less suited to chatbot assistance. This occupational heterogeneity supports the concept of a “jagged frontier” of AI chatbot costs and benefits (Dell’Acqua et al., 2023), warranting caution against extrapolating productivity estimates across tasks or occupations.

Relatedly, while RCTs often focus on tasks where AI chatbots are continuously applicable, the typical adopter in our data uses chatbots only every third to fourth day, diluting their overall time savings accordingly. On days when they do use the tools, however, average time savings still amount to just 6–7%, substantially below experimental estimates. Limited applicability may again be an important explanation: even on usage days, adopters spend only 5–6% of their work hours actively using AI chatbots (see Table E.10).⁴³

Second, reported time savings are significantly higher when employers actively encourage chatbot use. For example, marketers and software developers whose employers promote AI chatbots report time savings of approximately 7%. In this sense, RCTs may capture the productivity effects of a well-coordinated and carefully managed adoption of these tools. While such estimates provide valuable insight into the workplace potential of AI chatbots, our survey indicates that many users do not operate under similarly favorable conditions. This highlights the need for caution when extrapolating productivity benefits from controlled settings onto the broader economy.⁴⁴

⁴³The time use estimates should be interpreted with caution, as we might not expect a 100% active utilization rate even in tasks where AI chatbots are directly applicable. For example, in the writing task studied by Noy and Zhang (2023), treated workers spent on average 3 minutes before pasting in a large block of text—presumably generated by ChatGPT. The workers then spend an additional 14 minutes manually editing and refining the output. In our data, adopters spend an average of 2.2% of their total work hours using AI chatbots—less than the time savings they report. This pattern is also consistent with Noy and Zhang (2023), where treated workers saved 10 minutes on average despite only 3 minutes of initial tool use.

⁴⁴Another source of discrepancy between our time savings estimates and those from RCTs lies in the parameter of interest: whereas experiments typically identify Average Treatment Effects (ATE) for the study population, our estimates reflect self-reported time savings among users—closer to an Average Treatment Effect on the Treated (ATT). However, if workers adopt chatbots based on their expected individual benefits—such that users tend to derive greater gains than non-users—the modest time savings reported in our survey appear even more stark when contrasted with the larger effects found in RCTs.